Juneyoung Kim
Independent researcher, Seoul, Republic of Korea
lpaiu.cs@gmail.com

30 September 2026

The Editors
*Physica Scripta*

Dear Editors,

Please consider *"Finite-frequency identifiability of relaxation responses in free-fall timing"* as a Research Paper.

The manuscript addresses a methods question: when does a response pole remain identifiable after finite-frequency sampling and the fitting of nuisance parameters? Its scope connects mathematical and statistical physics with measurement and astrophysical inference.

**Proven.** We combine the classical real-interpolation boundary with an explicit nuisance-projected separation test and singular-value bounds. Exact controls distinguish a surviving response from one erased by projection or arbitrarily attenuated despite full rank. A positive-spectrum comparator requires an independently established relaxation-rate gap. The paper credits the established interpolation and estimation ingredients rather than claiming them as new general mathematics.

**Imported from prior work.** A stored PSR J0337+1715 analysis provides a conditional negative case: flexible comparators remove most of the reference information, an auxiliary drive fails a physical phase condition, and numerical derivative sensitivity prevents an empirical SEP interpretation. **Conjectural.** The remaining physical and statistical premises are stated explicitly; no relaxation detection or new empirical bound is claimed.

The main article contains the method, proofs, exact controls and case-study limitations. Supplementary Information preserves the detailed earlier audits and physical calculations. Supplementary Code reproduces the controls and compact conditional inference without the large private runtime arrays.

An earlier version was declined at editorial assessment by General Relativity and Gravitation, without referee reports. The manuscript is not published or under consideration elsewhere. There is no external funding or competing interest. Substantive AI assistance is disclosed in Methods and Acknowledgements, with author responsibility and verification limits stated.

Sincerely,

Juneyoung Kim